

G03 / ASSESSMENT

G03 GATE A

Standalone Zero-to-Job-Ready Data Engineer 2026

Study, practise, check and explain.

A complete learning route

Start with the lesson video. Read the worked explanation beside it, pause to reproduce the example, then solve the supplied task. The books contain your study notes; keep your own work and results as evidence.

Use the protected answers after saving your first attempt. A changed retry shows what you can now do independently. There is no forced timer during learning.

One course. M00-M19. 264 lessons and 50 project phases.

Contents

Tap a topic or use the PDF bookmarks. These numbers are the actual pages in this edition.

Read this first	3
YOUR STANDALONE GATE BOOK	3
Independent Python + Linux + Git checkpoint	3
Scenario	3
Required engineering proof	3
Attempt discipline	3

Read this first

YOUR STANDALONE GATE BOOK

Independent Python + Linux + Git checkpoint

Assessment rule: Save the current attempt before opening its review. Fresh Gate B stays closed until explicitly assigned after repair.

PROTECTED ASSESSMENT. Attempt only after M04, M05 and M06 are complete. Keep reviews and Gate B closed.

Scenario

A small operations team sends `orders_gate_a.csv`. Build a repository named `g03-order-quality` inside your Linux/WSL home workspace. The program must validate the input, reject bad rows with reasons, aggregate accepted amount by region, write deterministic JSON, and be runnable through a Bash wrapper.

Required engineering proof

1. Create `.venv`, `.gitignore`, `.env.example`, `README` and a clear project structure.
2. Preserve the raw source. Do not overwrite it while cleaning.
3. Python: reject missing `order_id`, duplicate `order_id`, invalid numeric amount and invalid status outside `Completed/Pending/Cancelled`. Save rejected records + reason.
4. Aggregate `Completed` accepted amount by region and include raw/accepted/rejected row counts in JSON.
5. Validate `raw = accepted + rejected` and accepted key uniqueness.
6. Bash wrapper accepts source path as argument; missing file must return non-zero.
7. Git: meaningful commits; a feature branch; one deliberate `README` conflict resolved by intent; one faulty commit reversed with `git revert`.
8. Remote mechanics: push/fetch using GitHub if actually available; otherwise local bare remote plus explicit `OFFLINE` label. Never fabricate GitHub evidence.
9. Clean clone into a different folder. Follow `README` only. Recreate `venv`, run wrapper, reproduce JSON and controls.
10. Technical English: 5-7 minute defense covering environment boundary, data validation, Git state, one failure, one repair and reproducibility.

Attempt discipline

Save the first attempt before opening Review A. A working final result with no saved reasoning/failure evidence is incomplete. Do not open Gate B until you have completed review/repair.